AI-Driven Fabric Defect Detection & Monitoring System for the **Apparel Industry** in Sri Lanka

Proposal Presentation

SLIIT
FACULTY OF COMPUTING

INTELLIGENT FABRIC CAPTURE, ANOMALY PRE-SCREENING, AND QUALITY ANALYTICS MODULE.

Specialization:
SOFTWARE ENGINEERING

Problem Definition & Shortcomings

- *Continuous fabric capturing generates huge volumes of redundant frames.*
- *Traditional systems capture every frame, Increasing latency and cost.*
- *No Intelligence at Capture Stage.*

Proposed Solution

1. **Motion-Driven Frame Capture**
   (Frames captured only when fabric motion is detected)
2. **Unsupervised Anomaly Pre-Screening**
   (Autoencoder learns normal fabric behavior and flags only irregular frames)
3. **Frame-Level Quality & Routing**
   (Each suspicious frame is scored for quality and routed intelligently)

Existing Research	Gap/ Limitations	Proposed System
Stojanović et al., 2001 – *Real-Time Vision-Based System for Textile Fabric Inspection*	Produces redundant data which is optimized only for roll inspections.	Introduces encoder-driven capture mechanism
Shiferaw & Yao, 2024 – *Autoencoder-Based Unsupervised Surface Defect Detection*	Applied only after full-frame acquisition, lacking edge-level real-time anomaly screening.	Introduce lightweight autoencoder to generate early anomaly scores + heatmaps during capture.
Raheja & Kumar, 2013 – *GLCM/Gabor Fabric Texture Analysis*	Used only for classification	Compute texture-density metrics (entropy, contrast) at capture layer

Key Domains of Knowledge Utilized

- **Computer Vision**
  Real-time image acquisition, preprocessing, and frame-level analysis using OpenCV for fabric inspection.
- **Edge AI / Embedded Machine Learning**
  Deployment of lightweight models (TensorFlow Lite) on Raspberry Pi for real-time anomaly pre-screening.
- **Signal Processing & Mechatronics Integration**
  Use of encoder pulse signals to synchronize frame capture with fabric movement for spatial consistency.
- **Unsupervised Learning**
  Autoencoder-based anomaly detection to identify irregular fabric regions without labeled datasets.
- **Software Engineering**
  Design of a multi-stage, real-time pipeline architecture with modular components and inter-process communication.

Proposed Technologies

- **Raspberry Pi 4 Model B + Camera Module 3**
  Low-cost edge device for real-time fabric capture and processing.
- **Python + OpenCV**
  Core implementation for frame acquisition, processing, and visualization.
- **TensorFlow Lite**
  Lightweight inference engine for deploying the anomaly autoencoder model on edge.
- **Encoder-Based Triggering Mechanism**
  Hardware-assisted frame capture based on fabric displacement (pulse-driven sampling).

Evaluate / Success Criteria

- **Spatial Sampling Accuracy**
  Frames captured at fixed physical intervals (~12 cm) using encoder pulses.
- **Real-Time Performance**
  System operates at ~28–30 FPS under edge constraints.
- **Data Reduction Efficiency**
  Reduces redundant frames and filters ~4–6% unusable frames before downstream processing.
- **Anomaly Detection Effectiveness**
  Successfully differentiates normal vs irregular frames using Frame Irregularity Score (FIS).
- **Quality Filtering Accuracy**
  Ensures only visually valid frames (based on FQS) are forwarded to fog layer.

Data Availability & Ethical Clearance

Data Sources

- Real-time fabric images captured using Raspberry Pi Camera Module 3
- Fabric samples from roll-based and cut-piece inspection setups
- Generated anomaly and quality metadata (FIS, FQS)

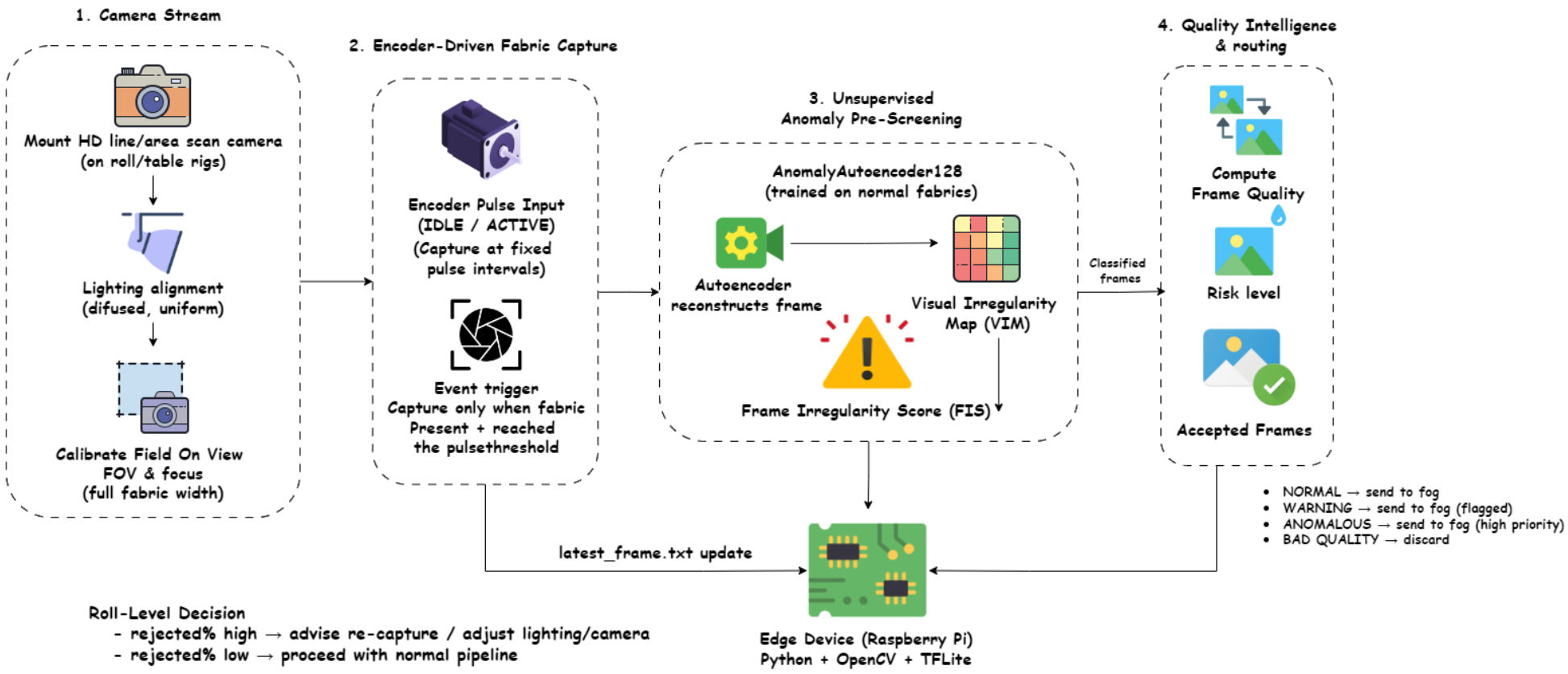
Ethical Compliance

- No personal or sensitive human data is collected
- System operates entirely on industrial fabric data

Real World Usage

- **Pre-Seam Fabric Inspection**
  Early detection of fabric irregularities before cutting or production.
- **Small & Medium Apparel Factories**
  Cost-effective alternative to expensive industrial inspection systems.

System Overview



EDGE-LEVEL FOG COMPUTING & IMAGE ENHANCEMENT

Specialization:

Software Engineering

Edge Level Fog Computing & Image Enhancement

Problem Definition & Shortcomings

- Fabric inspection images are affected by **poor lighting, noise and motion blur**
- Existing preprocessing methods use **fixed enhancement techniques (not adaptive)**
- Lack of **pattern-aware processing** (patterned vs non-pattern fabrics)
- Traditional methods struggle to **preserve texture while enhancing quality**
- Frame-level predictions are **unstable due to noise and defe**

Proposed Solution

- Adaptive Fabric-Aware Enhancement
- Lightweight Pattern Classification (CNN Models)
- Real-Time Edge Processing
- Quality-Aware Enhancement Pipeline

Feature	Proposed System	Existing Systems
Real-Time Processing	Yes (Edge-based, ~22–25 FPS)	Limited
Adaptive Enhancement	Yes (Pattern + Quality based)	No (Fixed methods)
Pattern Awareness	Yes (CNN-based classification)	No
Texture Preservation	High (edge-preserving methods)	No
Edge Deployment	Yes (Raspberry Pi)	Mostly Cloud / Offline
Computational Efficiency	Optimized (TFLite models)	High or inefficient
Integration with Detection	Yes (Improves downstream accuracy)	Limited

Key Domains of knowledge utilized

- Machine Learning / Deep Learning
- Computer Vision
- Edge Computing
- Software Engineering

Latest Technologies utilized

- EfficientNet-Lite0 & EfficientNet-B0 (CNN Models)
- TensorFlow Lite (TFLite)
- OpenCV (Image Processing Library)
- Next.js (Frontend Dashboard)

Evaluation/Success Criteria

- **Model Validation:** Accuracy, Precision, Recall, F1-score for: Pattern vs Non-pattern classification Pattern type classification (stripe, check, floral, geometric)
- **System Testing:** End-to-end pipeline validation (Capture – enhancement – defect detection)
- **Image quality improvement -** Improvement in brightness, contrast, and sharpness, Visual clarity before vs after enhancement , Noise reduction effectiveness
- **Expected Outcome:** ~92–94% pattern classification accuracy, ~22–25 FPS real-time processing

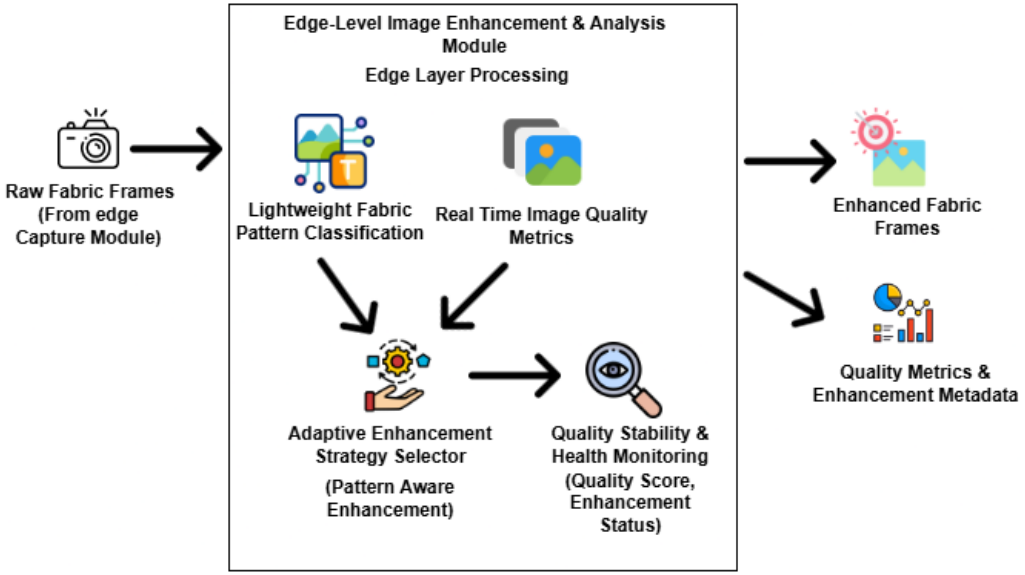
Data Availability

- **Non-Patterned Fabric dataset –** Manually Collected
- **4 Types of Pattern Fabrics –** Manually Collected

Ethical Consideration

- ❿ No personal or sensitive human data is collected
- ❿ System operates entirely on industrial fabric data

System Diagram



Real-World Application Scenarios of the Proposed System

- Pre-Seam Fabric Inspection in Apparel Manufacturing
- Small & Medium-Scale Textile Factories
- Defect Detection System Enhancement (Upstream Processing)

REAL-TIME FABRIC DEFECT DETECTION AND LOCALIZATION USING COMPUTER VISION AND IOT-BASED ROTARY ENCODER TRACKING.

Specialization:
SOFTWARE ENGINEERING

Proposed Definition & Shortcomings

- **Manual inspection is inefficient**
  Relies on human operators result slow, inconsistent, and error-prone
- **No precise defect localization**
  Existing systems detect defects but cannot map exact position on fabric
- **Delayed defect identification**
  Defects are often found after production, increasing material waste
- **High dependency on human intervention**
  Requires manual searching and decision-making

Solution Given

- **Real-Time Defect Detection -** YOLO-based AI detects defects instantly from fabric frames
- **Precise Localization –** Rotary encoder maps defects to exact position (±2 cm accuracy)
- **Automated Control System -** System auto-pauses when defect detected + resume control

Existing Approaches	Gap/Limitations	Proposed System
Manual Inspection	Slow, human errors, fatigue	Fully automated real-time system
Vision-only detection	No physical localization	Detection + exact position mapping
Delayed quality checks	Reactive approach	Real-time proactive detection
No machine control	No automatic response	Auto pause + operator control
Static models	Cannot adapt to new fabrics	Retrainable AI model

Key Domains of Knowledge Utilized

Computer Vision
Real-time image capture, preprocessing, and defect detection using YOLO and OpenCV

Embedded Systems & IoT
ESP32-based communication for real-time encoder tracking and hardware control

Mechatronics Integration
Rotary encoder + measuring wheel to synchronize fabric movement with frame capture

Machine Learning
Deep learning-based object detection and model retraining for new fabric patterns

Software Engineering
Design of real-time pipeline with modular backend, frontend, and hardware integration

Technologies Used

Hardware
- ESP32-S3 Development Board
- Rotary Encoder + Measuring Wheel
- NEMA 23 Stepper Motor
- TB6600 Stepper Motor Driver

Backend
- Python (FastAPI)
- OpenCV (image processing)
- Ultralytics YOLO (defect detection)
- Socket.IO (real-time communication)

Frontend
- Next.js (User Interface)
- WebSockets for real-time updates

Database
- MongoDB (defect history & analytic

Evaluate/Success Criteria

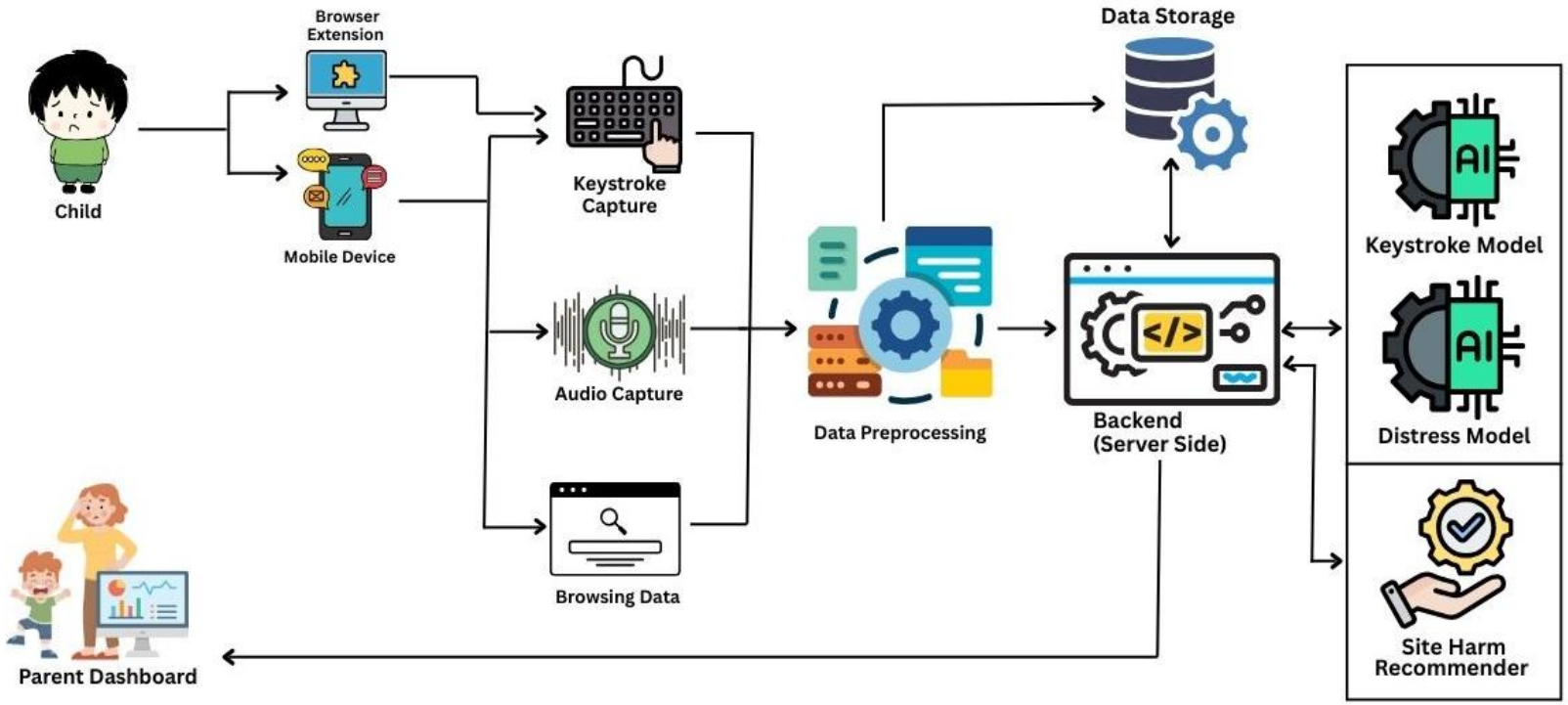
- **Detection Accuracy:** Achieves high accuracy (~94.7% mAP@50) across multiple defect types
- **Localization Precision:** Defects mapped with high accuracy (±2 cm) using rotary encoder.
- **Real-Time Performance:** Processes fabric frames continuously at ~5–10 FPS
- **System Responsiveness:** Automatically pauses on defect detection with manual resume control
- **Operational Efficiency:** Reduces manual inspection effort and improves quality control speed

Data Availability

- Real-time fabric images captured using camera system
- Fabric samples from inspection setup (various textures & defects)
- Generated defect metadata (type, position, confidence score)
- Historical defect records stored for analysis and monitoring

Ethical Consideration

- No personal or human-related data is collected
- System operates entirely on industrial fabric data
- No privacy or security risks involved
- Safe for deployment in manufacturing environments



Real-World Application Scenarios of the Proposed System

- Safe Search & Browsing
- Voice call risk detection
- Gaming Chat Intervention
- Messaging Application

Title of the Individual Component:

ML-POWERED PREDICTIVE ANALYTICS ENGINE FOR FORECASTING FUTURE FABRIC DEFECTS AND PRESCRIBING PREVENTIVE ACTIONS.

Specialization:

Software Engineering

Problem Definition & Shortcomings

- **no capability to predict future defect occurrences,** which li
ability of manufacturers to take proactive quality control measu
- **Current systems lack proper risk scoring mechanisms**, making it difficult to quantify the severity
- **No root cause identification**
- There is no effectively transformed into actionable insights for **decision-making and process improvement.**

Existing Approaches	Gap/Limitations	Proposed System
Defect detection only	No future defect prediction capability	Predicts future defect positions using LSTM.
No risk quantification	Cannot measure severity or likelihood of defects	Roll Risk Score (RRS) & Supplier Risk Score (SRS)
No root cause identification	No insight into why defects occur	Machine learning–based Root Cause Analysis (RCA)

Proposed Solution

- **Future Defect Sequence Prediction** – LSTM-based model forecasts future defect positions using historical defect sequences
- **Risk Scoring System** – Generates Roll Risk Score (RRS) and Supplier Risk Score (SRS) to quantify defect severity and likelihood
- **Root Cause Analysis (RCA) and Explainable AI Insights** – Machine learning model identifies possible causes behind defect patterns for preventive actions
- **Real-Time Analytics Dashboard**

Key Domains of Knowledge Utilized

- Machine Learning / Deep Learning
- Time-Series Analysis
- Data Analytics & Feature Engineering
- Explainable Artificial Intelligence (XAI)
- Software Engineering

ML-POWERED PREDICTIVE ANALYTICS ENGINE FOR FORECASTING FUTURE FABRIC DEFECTS ACTIONS.

Proposed Technologies

- **LSTM** (Sequential Prediction Model)
- **XGBoost** (Structured Risk Prediction)
- **Scikit-learn** (Encoding & ML utilities)
- **FastAPI** (Backend API Layer)
- **MongoDB** (Data Storage & Retrieval)
- **Next.js** (Dashboard Visualization)

Evaluate / Success Criteria

- **Model Evaluation:** Prediction accuracy of defect sequences, defects Pattern classification, Risk score consistency, RSA
- **System Testing:** End-to-end pipeline validation (Detection → Prediction → Risk → RCA), API response validation
- **Expected Outcome:** Accurate future defect prediction , Reliable risk scoring system, Explainable outputs for decision-making

Data Availability & Ethical Clearance

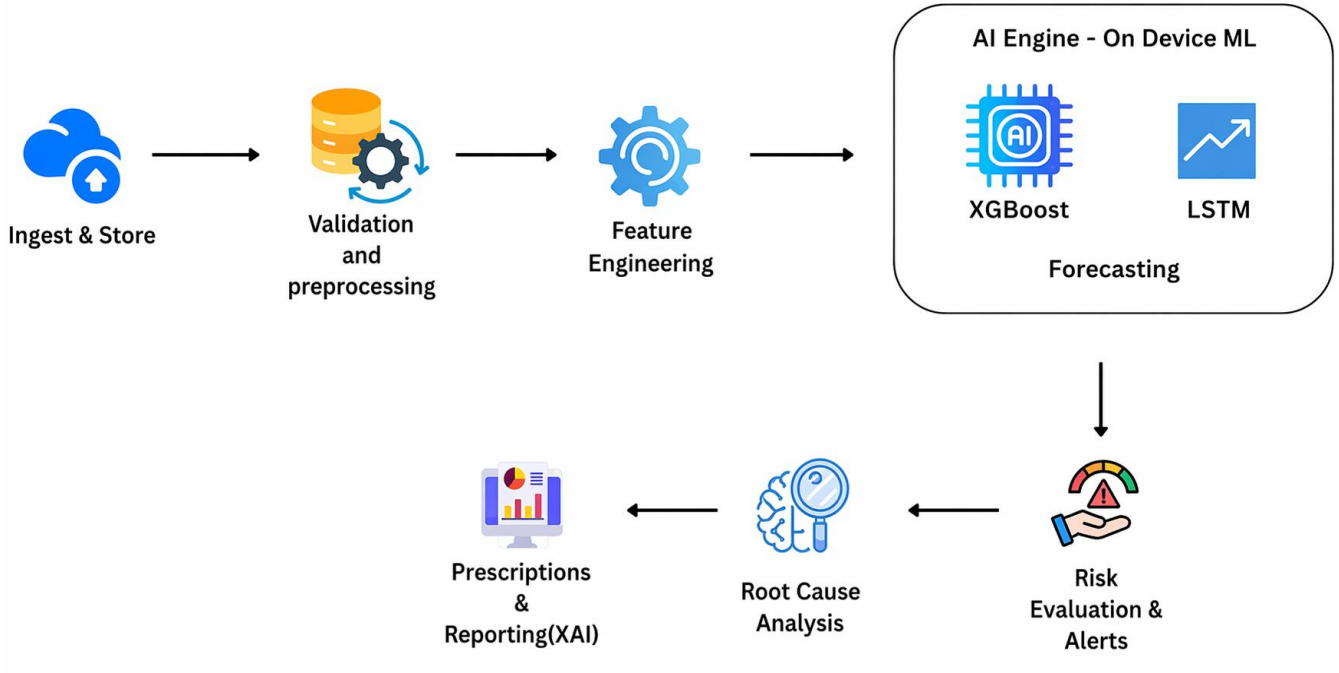
Data Availability

- Defect data from detection module (MongoDB)
- Sequence-based defect records (position + timestamp)
- Historical defect logs for training & testing

Ethical Compliance

- No personal or sensitive human data used
- System operates only on industrial fabric data

System Diagram



Real World Usage

- Pre-Seam Fabric Risk Prediction
- Supplier Quality Monitoring
- Early Defect Prevention in Production Lines
- Decision Support for Quality Managers
- Smart Manufacturing (Industry 4.0 Integration)

Commercialization Potential

- *AI-Powered automated fabric inspection system.*
- *Reduces manual errors, labor cost and fabric waste.*
- *Real-time quality monitoring.*

Business Model

- *Hardware + software integrated solution.*
- *Modular deployment (component-based).*
- *Licensing / industrial integration.*

Target Market

- *Garment manufacturers & textile mills.*
- *Apparel exporters (quality compliance).*
- *Industrial automation providers.*

Competitive Advantage

- *Event-based intelligent frame capture.*
- Early anomaly detection + quality filtering.
- Real-time defect detection & predictive analytics.
- Cost-effective compared to industrial systems.

Thank You!